

- Q1.** For the filter described by $y[n] = x[n] - x[n - 1]$ and the input sequence $x[n] = \{1, 1\}$:
- (a) Determine the impulse response, $h[n]$, i.e., the response due to the signal $\delta[n] = \{1\}$.
 - (b) Determine the response of the filter, $y[n]$, assuming zero initial conditions, i.e., $y[n] = 0$ for $n < 0$.
 - (c) Determine the Z-transform, $H(z)$, of the filter impulse response, $h[n]$.
 - (d) Determine the Z-transform, $X(z)$, of the input sequence, $x[n]$.
 - (e) Determine the Z-transform, $Y(z) = H(z)X(z)$, of the filter response, $y[n]$.
 - (f) From the Z-transform of the filter response, $Y(z)$, determine the filter response $y[n]$.
- Q2.** For the filter described by $y[n] = x[n] + 2x[n - 1] + x[n - 2]$ and the input sequence $x[n] = \{1, -1\}$:
- (a) Determine the impulse response, $h[n]$, i.e., the response due to the signal $\delta[n] = \{1\}$.
 - (b) Determine the response of the filter, $y[n]$, assuming zero initial conditions, i.e., $y[n] = 0$ for $n < 0$.
 - (c) Determine the Z-transform, $H(z)$, of the filter impulse response, $h[n]$.
 - (d) Determine the Z-transform, $X(z)$, of the input sequence, $x[n]$.
 - (e) Determine the Z-transform, $Y(z) = H(z)X(z)$, of the filter response, $y[n]$.
 - (f) From the Z-transform of the filter response, $Y(z)$, determine the filter response $y[n]$.
- Q3.** For the filter described by $y[n] = 0.8y[n - 1] - 0.2x[n]$:
- (a) Determine the first four non-zero samples of the impulse response of the filter, i.e., the response due to the signal $\delta[n] = \{1\}$.
 - (b) What is the value of $y[100]$ when $x[n] = \delta[n]$? [Hint, generalise the previous answer.]
 - (c) Determine the impulse response, $h[n]$, of the filter for all n .
 - (d) Determine the Z-transform, $Y(z)$, of the filter response, $y[n]$.
 - (e) Determine the Z-transform, $H(z)$, of the filter impulse response, $h[n]$.

- (f) Determine the impulse response, $h[n]$, from the transfer function, $H(z)$.
- Q4.** (a) What is the impulse response of the filter defined by $y[n] = 0.75x[n] - 0.25x[n-1]$? Is it a FIR filter?
- (b) What is the impulse response of the filter defined by $y[n] = 0.75y[n-1] - 0.25x[n]$? Is it a FIR filter?
- Q5.** (a) Write the sequence $\{1, 2, 3\}$ as a weighted sum of impulses.
- (b) Write the sequence $\{1, 2, 3\}$ as a weighted sum of impulses.
- (c) Write $1\delta[n] - 3\delta[n-2] + 5\delta[n-4]$ as a sequence.
- (d) Determine the convolution of the sequences $x[n] = \{1, 1, 1\}$ and $h[n] = \{1, 1, 1\}$?
- (e) Determine the convolution of the sequences $x[n] = \{1, 2, 3\}$ and $h[n] = \{1, 1\}$?
- (f) What is the extent of the convolution of the sequences $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $\{8, 7, 6, 5, 4, 3, 2, 1\}$?
- (g) How many multiplications are required for the convolution in the previous question?
- (h) If sequences of extents 100 and 50 are convolved, what is the extent of the result?
- (i) If sequences of extents 100 and 50 are convolved, how many multiplications are required?

Q6. Consider the sequence $x[n] = \{1, 2, 1, 3\}$ to be convolved with a sequence $h[n] = \{0.5, 0.5\}$.

- (a) Express the sequence $x[n]$ as a weighted sum of delayed impulses.
- (b) Sketch the lollipop (stem) plot for $x[n]$.
- (c) Express the sequence $h[n]$ as a weighted sum of delayed impulses.
- (d) Sketch the lollipop (stem) plot for $h[n]$.
- (e) Sketch the lollipop plot for $x[n]$ convolved with $h[n]$. Hint: expand the convolution of $x[n]$ with $h[n]$ given by

$$y[n] = \sum_{m=0}^{M-1} h[m]x[n-m]. \quad (1)$$

- (f) What is the Z-transform of $h[n]$?
 - (g) What sort of filter does this convolution perform?
- Q7.**
- (a) A 1024 point FFT is calculated for a signal sampled at 1 kHz. What is the spacing between frequency samples?
 - (b) A signal is sampled at 1 kHz for 1024 samples, zero-padded to a length 4096, and its spectrum calculated with a FFT. What is the spacing between frequency samples?
 - (c) A discrete spectrum calculated with a FFT has 512 frequency samples spaced by 1 kHz. What is the spacing of the samples of the corresponding discrete-time signal?
 - (d) What is the purpose of zero-padding when performing convolution with FFTs?
 - (e) A 100 sample sequence is to be convolved with a 150 sample sequence using FFTs. Determine the length of the output sequence and suggest a suitable FFT size.
 - (f) Calculate the DFT of the sequence $\{1, 2, 2, 1\}$.
 - (g) What is the purpose of applying a window function before using an FFT?